

Research papers

Changes of preferential flow in short-rotation eucalyptus plantations: field experiments and modeling

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ABSTRACT

As an important part of the hydrological cycle, preferential flow significantly improves infiltration properties and limits the ability of soil to store, filter, and buffer water. However, the effect of roots on the preferential flow infiltration process in forest ecosystems is still unclear. This study measured the preferential flow infiltration process in short-rotation eucalyptus plantations (four forest ages of 1, 2, 3, and 5 years) with the surface-positioned double-ring infiltrometer in south subtropical area, China. Overall, the cumulative infiltration (CI), initial infiltration rate (IIR), and stable infiltration rate (SIR) increased with forest age. The CI, IIR, and SIR of the 5-year-old eucalyptus plantation were the highest in four aged eucalyptus plantations. Correlation analysis showed that root biomass was positively and significantly correlated with CI ($p < 0.05$), IIR ($p < 0.01$), and SIR ($p < 0.01$), which showed that the preferential flow increased with root biomass. However, the correlations between total porosity (TP) and IIR, SIR, and CI were all not significant ($p > 0.05$). The dye coverage of the inner and outer ring tangents, which were used to observe the strength of lateral preferential flows, was significantly and positively correlated with CI ($p < 0.01$), IIR ($p < 0.01$), and SIR ($p < 0.01$). The dye coverage of the outer ring tangent was significantly and positively correlated with root biomass ($p < 0.05$). So, the lateral preferential flows depended on root biomass and increased with forest age. The Horton model, which had higher stability and accuracy than the Kostiaikov and Philip models, was the most suitable model to describe the preferential flow infiltration process in eucalyptus plantations. As important factors influencing the evolution of preferential flow in eucalyptus plantation, root biomass was used in predicting the parameters of the Horton model through regression analysis. Then, a unified model predicting the preferential flow infiltration process in short-rotation eucalyptus plantations was proposed and verified. This study achieved the quantitative measurement of preferential flow infiltration process, explored the importance of roots in the production of preferential flow channels, and found the lateral preferential flow played an important role in infiltration in forests. These results can provide a theoretical basis for improving the efficiency of water resource management in plantation, and optimizing the model of preferential flow infiltration process in plantation.

1. Introduction

Preferential flow is defined as the physical movement of solutes and water along certain pathways, while bypassing most of the soil porous matrix (Hendrickx and Flury, 2001; Beven and German, 2013), accounting for about 15%–85% of the soil total infiltration (Laine-Kanlio

et al., 2014; Zhang et al., 2018; Zhang et al., 2020). Preferential flow was an important part in the regional and global water cycle (Hirmas et al., 2018; Jiang et al., 2020). Preferential flow improves soil infiltration property, thereby reducing surface runoff and soil erosion (Allaire et al., 2011), increasing groundwater recharge, and slowing down soil degradation (Kan et al., 2020). Meanwhile, preferential flow shortens the

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contact time between water and soil and limits the ability of soil to store, filter, and buffer water (Alakukku et al., 2010; Laine-Kaulio et al., 2015), thus reducing the water and nutrient utilization rates of plants (Clothier et al., 2007; Luo et al., 2019) and increasing the risk of groundwater pollution (Weiler, 2017). Therefore, the study of preferential flow is essential to improving the utilization efficiency of water resource, comprehensively evaluating the benefits of soil and water conservation, and preventing and controlling groundwater pollution, which has been a global concern (Kulasekera et al., 2011).

The macropores, accounted for only a small proportion of soil (Beven and German, 2013; Moret and Arrue, 2007), are formed in many ways, such as root growth, chemical weathering, and drying-wetting and freezing-thawing cycles (Hardie et al., 2011; Li et al., 2020; Chen et al., 2021). Thereinto, root growth is considered as an important way forming macropores (Nespoulous et al., 2019). Both decaying and living roots can form preferential root channels or continuous macropores and promote the movement of water and solutes with slight resistance (Gaiser, 1952; Jiang et al., 2018; Liu et al., 2020). Root exudates enhance soil aggregation that form inter-aggregate macropores (Luo et al., 2018). Root channels account for 70%–100% of the total amount of macropores in forest soils and serve as main paths of water and solute preferential transport (Newman et al., 2004; Noguchi et al., 2017). The radii of developed forest roots are almost 10 times the canopy radius, and the development of forest roots lead to a wide distribution range and high degree differentiation of preferential flow in forest soils, and produced a higher lateral preferential flow (Lejeune et al., 2004; Ghestem et al., 2011; Liu et al., 2021) and infiltration rate (Neris et al., 2012; Mongil-Manso et al., 2021). To date, many researchers conducted a lot of qualitative studies on the relationship between forest roots and preferential flow by using the dye tracer technique, but mainly focusing on the relationship between the dye covers and root diameter (Luo et al., 2018), root length density (Zhang et al., 2015), and root weight density (Gang et al., 2017). The effect of root biomass on the lateral preferential flow developing, and the quantitative relationship between root biomass and preferential flow infiltration processes were still unclear.

As forest age increases, plant root channels become deeper, wider, and denser, facilitating preferential flow and increasing the permeability of forest soils (Tobella et al., 2014). Deuchars et al. (1999) measured soil infiltration rate by single ring infiltrometer to study the change in soil infiltration rate during rainforest regeneration after pastures were abandoned in the San Luis Valley in Central America. Their results showed that soil infiltration rate increased with forest age because the root activities in the forests increased soil porosity and decreased soil bulk density. Greenwood and Buttle (2014) and Archer et al. (2016) also found the saturated hydraulic conductivity was increased with forest age in southern Ontario and temperate forests, because root biomass and macroporosity increased. Kalhor et al. (2018) showed that forests and grasslands had positive effects on soil infiltration, because of increasing root biomass and total porosity (TP) in the hilly areas of the Loess Plateau, and the effects strengthened as planting age increased. Shi et al. (2021) found that plant age could significantly affect soil hydraulic properties, and root development significantly promoted soil infiltration by using meta-analysis. They proposed that the correlation between TP and infiltration was not significant, while the high correlation with non-capillary porosity. Above studies mainly focused on the changes of total infiltration with forest age, but not distinguishing the matrix infiltration and preferential flow. Some studies found the similar change of preferential flow with forest age. Cheng et al. (2014) and Ma et al. (2022) studied the vertical distribution characteristics of soil preferential paths in differently aged Chinese fir plantations and *caragana korshinskii* plantations by using the dye tracer method, respectively. Their results showed that the preferential flow path depths and dye covers increased with increasing forest age because of root mass and number increasing. So, the root system of forest continuously developed with the increase of forest age to increase the root biomass, and then affecting the preferential flow path and promoting the total infiltration and preferential

flow. However, previous studies were confined to qualitatively analyze dye covers by using the dye tracer method. Quantitatively analysis and predicting model with considering root characteristics at different forest age were lacked on studying preferential flow.

Many methods have been developed and applied for studying preferential flow, such as dye tracer technique (Nespoulous et al., 2019), time domain reflectometry (Oberdörster et al., 2010), 3D X-ray computed tomography (CT, Sammartino et al., 2012), magnetic resonance imaging (Votrubová et al., 2003), and ground-penetrating radar (Di Prima et al., 2020). The dye tracer technique is the most commonly used method in studying preferential flow and allows the direct observation, evaluation, and analysis of the path quantity and distribution of preferential flow (Jiang et al., 2019). Zhang (2017) and Zhang et al. (2019) proposed a surface-positioned double-ring infiltrometer for determining the preferential flow infiltration process and measuring difference in total infiltration and matrix infiltration in adjacent areas. They observed the distribution of preferential paths and successfully measured the preferential flow infiltration process in 50-year-old *Robinia pseudoacacia* in the Loess Plateau, China. Compared with other methods, Zhang's method is relatively economic, simple, and easy to operate and can accurately quantify the infiltration processes of preferential flow (Tao et al., 2019).

In the Guangxi Zhuang Autonomous Region, eucalyptus plantations constitute 34.91% of the total plantation area, and eucalyptus contributes to 49.97% of harvested timber stock volume (Zhang and Wang, 2021). Owing to the short-rotation of eucalyptus plantation (5–7 years), eucalyptus plantations require continuous management while release active allelopathic substances, which inhibit the growth of understory vegetation resulting in sparse understory vegetation (Engler et al., 2016; Yang et al., 2016; Zhou et al., 2018; Yao et al., 2021). Thus, eucalyptus plantations hydrological cycles were heavy disturbed and causing problems in water environments (Deng et al., 2020). Therefore, studying on preferential flow and its control mechanism in eucalyptus plantation is essential to investigate the water ecological security in the region. Simultaneously, the measurement of preferential flow in eucalyptus plantations is difficult in the Guangxi Zhuang Autonomous Region due to the large number of mountains and inconvenient transportation (Han and Song, 2020). A unified model describing the preferential flow infiltration process in short-rotation eucalyptus plantations is urgently needed, which can improve measurement efficiency and reduce experiment cost. Thus, combined with the findings of previous studies, a hypothesis was proposed: the growth of root system with forest age could promote the forming of lateral preferential flow in eucalyptus plantations and the preferential flow infiltration process. From this, we present the following objectives: (1) to quantify the preferential flow infiltration process in four aged eucalyptus plantations; (2) to assess the effect of root biomass and TP on preferential flow; and (3) to establish a unified model with considering root biomass for predicting the preferential flow in short-rotation eucalyptus plantations.

2. Materials and methods

2.1. Study area

Four aged eucalyptus plantations at the state-owned Dongmen Forest Farm (107° 15'–108° 00' E, 22° 17'–22° 30' N) in Chongzu city in China were selected as study objects (Fig. 1a). The annual rainfall and evaporation are 1100–1300 mm and 1192–1724 mm, respectively. Rainfall events mainly occur from June to August, accounting for 51.03% of the annual rainfall. The average annual temperature is 21 °C. The soil is lateritic red soil with pH of 4.0–6.0. The soil has a clear development differentiation profile and good hydrothermal conditions.

2.2. Experiment design

Four *Eucalyptus urophylla* × *E.grandis* plantations planted in May in

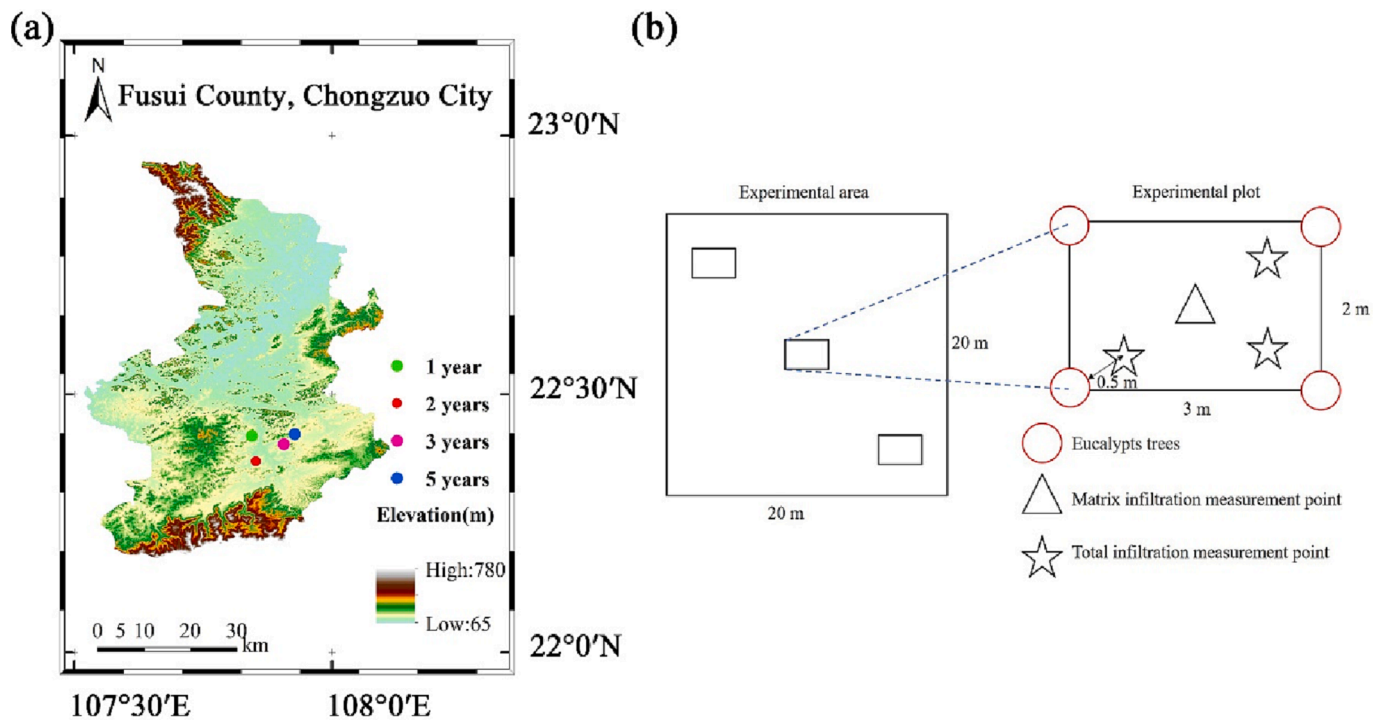


Fig. 1. Location of the experimental area and specific location of measurement points. (a) Location of the experimental area; (b) Specific location of the measurement points of matrix infiltration and total infiltration.

2020, 2019, 2018, and 2016 were selected as experimental forests (Table 1). And the gradient of forest age represented the different growth stage of eucalyptus plantations in short rotation (Ouyang et al., 2021). In each plantation, a 20 m × 20 m experimental area was set, and three experimental plots (2 m × 3 m) were set along the diagonal, randomly. The four corners of each experimental plot consisted of four eucalyptus trees as shown in Fig. 1b. The measurement place of matrix infiltration was the center of each experimental plot. Three of four adjacent eucalyptus were selected as the sample trees, namely nine sample trees in each experiment plot. Given that the root systems of eucalyptus plantations are mainly distributed within 1 m² from the center of the trunk (Viera et al., 2012; Laclau et al., 2013), a flat and grassless place that was 0.5 m away from the trunk was selected as the measurement point of total infiltration. Three matrix infiltration measurements and nine total infiltration measurements were performed in each plot. All eucalyptus plantations were the first generation of seedling forests. The cluttered shrubs or weeds shoveled were cut every 1 year. The spaces and intra-spaces in all plantations were 2 m × 3 m. Eucalyptus have been planted in the four experimental areas since 2001 and our experiments were performed in August 2021.

2.3. Soil samples analysis and root biomass estimation

Three points were randomly selected diagonally in each eucalyptus plantations plots to excavate soil profiles. Because the soil layer below 60 cm of lateritic red soil is a parent material layer and the depth of soil

subsoiling was mostly 50–60 cm in eucalyptus plantation (Fu and Li, 1993; Santos and Reichert, 2022), soil samples of 0–60 cm soil layer were collected to use in laboratory analysis in this study. Three replicate samples were collected at three layers (0–20 cm, 20–40 cm, and 40–60 cm) of each soil profile for measuring soil water content (SWC), bulk density (BD), soil particle density (ds), TP, and soil texture. These indicators of the whole soil profiles were counted with mean values of the

Table 2

Soil properties and diameter at breast height of the sample trees (mean ± SD) in the eucalyptus plantations.

Soil properties	Ages of eucalyptus plantation (year)			
	1	2	3	5
Initial soil water content (%)	21.62 ± 0.01b	18.27 ± 0.01c	24.35 ± 0.01a	16.85 ± 0.01d
Bulk density (g cm ⁻³)	1.07 ± 0.05c	1.32 ± 0.03a	1.08 ± 0.02c	1.24 ± 0.05b
Soil particle densities (g cm ⁻³)	2.77 ± 0.02a	2.68 ± 0.03a	2.73 ± 0.04a	2.65 ± 0.04a
Soil total porosity (%)	61.23 ± 2.00a	50.31 ± 1.19c	60.38 ± 0.80a	53.41 ± 1.77b
Diameter at breast height (cm)	5.94 ± 1.12c	6.85 ± 1.41c	13.79 ± 1.00b	15.11 ± 1.38a

Note: Different lowercase letters indicate significant differences in soil properties among differently aged eucalyptus plantations at the 0.05 level. The measured values are represented as means ± standard deviation.

Table 1

Basic information of the experimental plots.

Forest age (year)	Planting year	Location	Elevation (m)	Plant height (m)	Canopy density (%)	Thickness of litter (cm)
1	2020	107°50'39"E, 22°25'7"N	95.60	5.00	61.30	3.75
2	2019	107°51'10"E, 22°22'10"N	127.43	5.70	61.24	4.50
3	2018	107°54'23"E, 22°24'8"N	128.00	14.13	56.50	5.50
5	2016	107°55'39"E, 22°25'20"N	151.00	18.88	45.68	14.88

three soil layers (Table 2). The soil samples for measuring SWC were sealed immediately in airtight aluminum cylinders, and SWC was measured by the oven drying method in laboratory. Soils were measured for calculating BD were collected by the cutting ring (100 cm³), and oven-dried at 105 °C until constant weight. For the other analyses, the soil was sampled with a small shovel, air-dried, removed roots and passed through a 2-mm sieve in the laboratory. TP was calculated using Eq. (1), and soil particle density was measured by the pycnometer method.

$$TP = (1 - BD/ds) \times 100 \quad (1)$$

where TP is total porosity, %; BD is bulk density, g cm⁻³; and *ds* is soil particle density, g cm⁻³. Soil texture was determined using a Mastersizer 3000 analyzer (Malvern Instruments). The ranges of sand, silt, and clay in the four plantations were 11.01%–29.39%, 51.43%–59.89%, and 14.57%–37.56%, respectively. According to the USDA texture classification, the forest soils of the four plots were loam soil.

Wang et al. (2015) arranged 15 sample plots of eucalyptus plantation by using the grid method (100 km × 100 km) in Guangxi Zhuang Autonomous Region. Each sample plot had three replicates (20 m × 50 m). The height, diameter at breast height, and crown breadth of a tree in each duplicate plot were measured. A total of 18 eucalyptus trees with different ages and diameters at breast height were selected outside each plot for the measurement of root biomass. The distribution level of diameter at breast height was considered during measurement. Then, the optimized allometric model (Eq. 2) was formulated on the basis of the dataset of root biomass and diameter at breast height of the eucalyptus plantations and used in estimating root biomass. The root biomass of each sample tree was estimated with the optimized allometric model (Wang et al. 2015).

$$W_R = 0.06D^{1.771}R^2 = 0.85 \quad (2)$$

where W_R is root biomass, kg and *D* is diameter at breast height, cm.

In this study, the diameter at breast height (1.3 m above ground level) of the sample trees was measured with tape measure (Table 2).

2.4. Measurement and calculation method of preferential flow

2.4.1. Measurement method of total infiltration and matrix infiltration

A surface-positioned double-ring infiltrometer (inner ring diameter of 30 cm, outer ring diameter of 60 cm, and both height of 8 cm) was used to measure total infiltration and matrix infiltration (Zhang et al.,

2017a, 2019). As shown in Fig. 2a, a flat measurement point was selected and carefully cleaned the litter for total infiltration measurement. The double-ring infiltrometer was gently hammered into the soil to a depth of about 1 cm by using a leveling ruler and a rubber hammer, while ensuring that the rings remained horizontal. The method requires the water level of the inner and outer rings keeping consistent. The water of outer ring prevents the water of inner ring lateral seepage, so that keeping a one-dimensional vertical infiltration process in the inner ring (Zhang and Li, 2020). Soil was stacked and compacted at the inner and outer walls to prevent water leakage from the bottoms of both rings. Two 5 cm × 5 cm cardboards were placed in the inner and outer rings to buffer water flow and prevent damage for the surface soil structure. At the beginning of the experiment, the 4 g L⁻¹ methylene blue solution was poured into the inner ring and the clean water was poured into the outer ring at the same time. Based on Zhang's experience, the heights of infiltration rings were set as 8 cm. The measurement of soil matrix infiltration needed hammering into the soil of 1 cm and covering 3 cm-thick soil-sand mixture, thus the maximum measurement water level of soil matrix infiltration was only 4 cm. The initial water level of soil total infiltration kept be consistent with soil matrix infiltration. Simultaneously, many researches also set a water level of 4 cm when using surface-positioned double-ring infiltrometer (Zhang, 2017; Huang et al., 2018; Tao et al., 2019; Zhang et al., 2019) and traditional double-ring infiltrometer (Bodhinayake et al., 2004; Wu et al., 2021). Therefore, both the initial water levels of soil total infiltration and matrix infiltration were 4 cm depth. A stopwatch was used to record the time durations of each 1 cm water infiltration in inner ring. Throughout each experimental process, water was appropriately added into the outer ring to maintain the water levels in the inner and outer rings consistent. Other details of the measurement of total infiltration could be found in previous studies (Tao et al., 2019; Zhang et al., 2019).

The surface-positioned double-ring infiltrometer with covering materials was used in measurement of soil matrix infiltration. Perroux and White (1988) indicated that the saturated hydraulic conductivity of a covering material for sheltering preferential flow must be higher than that of the soil matrix in a tested plot to ensure that the infiltration process was controlled by the soil matrix. And Yuan (2018) found that the soil-sand mixture as covering materials, whose saturated hydraulic conductivities were higher than that of the soil matrix, could well shelter preferential flow. Among them, the higher the proportion of soil, the better shielding effect on preferential flow. The saturated hydraulic conductivity of soil-sand mixtures with different proportions (5:0, 4:1, 3:2, 1:1, 2:3, 1:2, 1:3, and 0:5) were pretested for the identification of

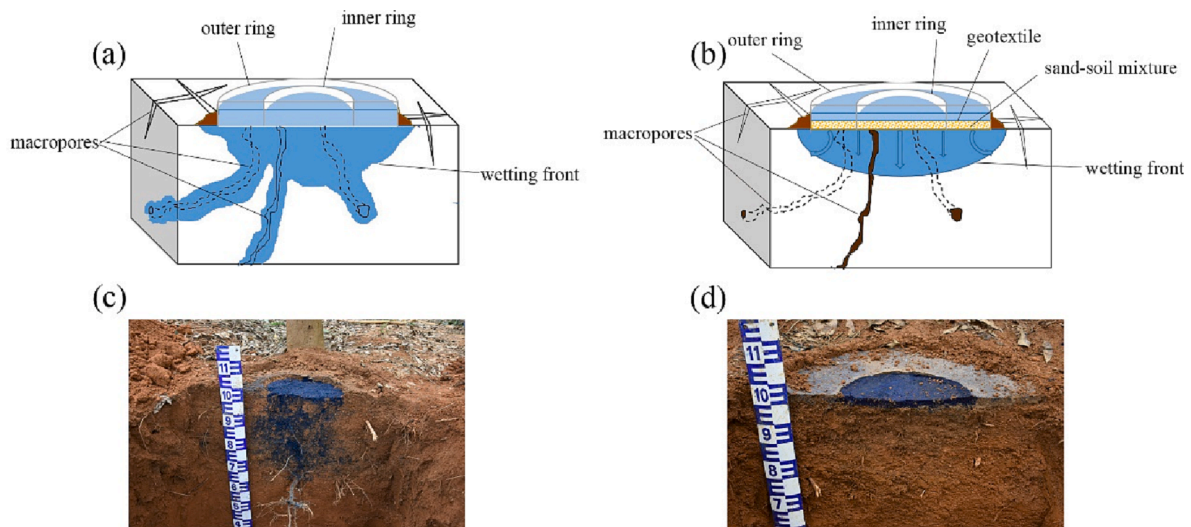


Fig. 2. Diagram of measuring infiltration process of (a) total infiltration and (b) matrix infiltration, and the soil profile after finishing (c) total infiltration and (d) matrix infiltration.

optimal soil-sand mixtures. A series of laboratory experiments were conducted and obtained that the saturation hydraulic conductivities of topsoil (0–20 cm; 216 mm h⁻¹, 128 mm h⁻¹, 121 mm h⁻¹, and 98 mm h⁻¹) of 1-, 2-, 3-, and 5-year-old eucalyptus plantations and that of 1:2 and 2:3 soil-sand mixtures (238 mm h⁻¹ and 112 mm h⁻¹). According to the test results, 3 cm-thick soil-sand mixtures (1:2) were uniformly spread over the areas of inner rings and the areas between inner and outer rings in the 1-, 2-, and 3-year-old eucalyptus plantations, and 2:3 soil-sand mixtures were used in the 5-year-old eucalyptus plantation (Fig. 2b). To buffer the water flow, geotextile and two 5 cm × 5 cm cardboards were placed on the surface of soil-sand mixture. Subsequent measurement procedures were same with those used in total infiltration. Using soil-sand mixture effectively to block preferential flow ensures the accuracy of matrix infiltration measurement (Yuan, 2018; Zhao et al., 2022).

When no change in infiltration time was observed after three times of measurement, the infiltration process was considered stable infiltration. Commonly, the time of measurement is 1.5 h. The infiltration rate was calculated with Eq. (3):

$$i = \Delta I / \Delta t \times 600 \quad (3)$$

where, i denotes infiltration rate, mm h⁻¹; ΔI is the cumulative infiltration in the inner ring during Δt time period, cm; and 600 is the conversion coefficient.

After 24 h of finishing the measurement of total infiltration and matrix infiltration, 1 m × 1 m soil profiles were excavated (Fig. 2c and d). A series of vertical soil profiles along the tangents of the outer ring, the inner ring, one-fourth of the inner ring, and half of the inner ring were respectively excavated. Then, a series of horizontal soil profiles at depths of 5, 10, 20, and 30 cm from the soil surface according to the profile of the half outer ring were excavated. The soil profiles were treated with Photoshop CS6 (Adobe Systems Incorporated, USA) and ImageJ (Wayne Rasband, National Institutes of Health, Maryland, USA) for the quantification of the dyeing area (Yan and Zhao, 2016). The images were geometrically corrected with Photoshop CS6. In accordance with the inner ring width and height of the soil profile, each image was cropped to 30 cm × 70 cm. After the saturation, brightness, gray-scale, and threshold of the processing pictures were adjusted, the color images were converted to black-and-white binary images, which were used in calculating the proportion of the stained area to the entire profile, with ImageJ.

2.4.2. Calculation principle of preferential flow

Owing to the small spatial variability of matrix infiltration in adjacent area (Das Gupta et al., 2006; Zhang et al., 2017a), Zhang (2017) and Zhang et al. (2019) proposed a principle for estimating the preferential flow infiltration process. Soil infiltration with many macropores consists of two parts: soil matrix infiltration and preferential flow. And soil matrix infiltration in the preferential flow measurement area can be replaced by that in the adjacent area. Therefore, the formula for calculating the preferential flow is:

$$i_{2p} = dQ_{2p}/dt = dQ_2/dt - dQ_{1m}/dt = i_{2t} - i_{1m} \quad (4)$$

where Q_{1m} represents the matrix infiltration of position 1 (mm) and Q_{2m} is the matrix infiltration of position 2 (mm); i_{2p} is the infiltration rate of preferential flow of position 2, mm h⁻¹; i_{2t} is the infiltration rate of total infiltration of position 2, mm h⁻¹; and i_{1m} is the infiltration rate of matrix infiltration of position 1, mm h⁻¹.

Other details of the calculation principle of preferential flow could be found in the study of Zhang et al. (2019). After the measurement of soil total infiltration and soil matrix infiltration in adjacent areas, the preferential flow infiltration process can be obtained with Eq. (4).

2.4.3. Fitting model of preferential flow infiltration process

The Horton, Kostikov, and Philip models were used to fit the

measured preferential flow infiltration process. The Horton model considers the effect of soil surface conditions on infiltration property when describing the rainfall infiltration process (Horton, 1941).

$$i = i_f + (i_0 - i_f)\exp(-nt) \quad (5)$$

where i denotes infiltration rate, mm h⁻¹; t is infiltration time, h; i_f represents stable infiltration rate, mm h⁻¹; i_0 represents initial infiltration rate, mm h⁻¹; and n is infiltration decay factor.

The Kostikov model is the most commonly used empirical model (Kostikov, 1932). Its parameters have strong elasticity.

$$i = at^{-b} \quad (6)$$

where a and b are fitting parameters and a and b describe the infiltration rate at t -time.

The Philip model is a classical physical model for simulating infiltration (Philip, 1957). However, the fitting deviation of the Philip model for long-time infiltration is large.

$$i = 0.5St^{-0.5} + A \quad (7)$$

where S is the sorptivity (mm h^{-0.5}) and A is conceptually the stable infiltration rate (mm h⁻¹).

In addition, the expression of CI was obtained according to the functional relationship (Eq. 8) between infiltration rate and CI.

$$CI = \int_0^t i(t)dt \quad (8)$$

where CI is cumulative infiltration, mm.

2.4.4. Unified model of preferential flow

The establishment of a unified model for predicting the preferential flow infiltration process requires four steps: (1) preferential flow infiltration process of six of the nine sample trees in each plantation were randomly selected and fitted by the Horton, Kostikov, and Philip models. And determination coefficients (R^2) were compared on the basis of the preferential flow infiltration process fitted by the three models for the identification of the model can characterize the preferential flow infiltration process; (2) the measured values of cumulative infiltration (CI), initial infiltration rate (IIR), and stable infiltration rate (SIR) were compared with the analytical values estimated by the three models, and the model with the highest fitting accuracy was selected as the basic model; (3) the correlation between infiltration process and root biomass was analyzed, and root biomass was used as an independent variable for predicting the original parameters of the basic model through regression analysis; (4) a unified model was obtained for predicting the preferential flow infiltration process, and the accuracy of the unified model was verified by other three trees.

2.5. Statistical analysis

In this study, IIR is the average of the first 3 min of infiltration, SIR is the average of the last three infiltration rates, and CI is cumulative infiltration of preferential flow within 1.5 h. All data were presented as mean ± SD, and the Shapiro-Wilk test was used to verify the data conformed to the normal distribution. One-way analysis of variance was used to analyze differences in soil properties, root biomass, CI, IIR, and SIR of eucalyptus plantation of different ages. The regression analysis was used to derive the regression equations predicting the model parameters. The correlation analysis was used to analyze the correlation among dye coverage of soil profiles along the tangents of inner and outer rings, root biomass, TP, IIR, SIR, and CI.

3. Results

3.1. Preferential flow in eucalyptus plantations of four ages

The total infiltration and matrix infiltration of four eucalyptus plantations are shown in Fig. 3. The preferential flow infiltration process (Fig. 4) was obtained with Eq. (4). The results showed that IIRs were $4314 \pm 2708 \text{ mm h}^{-1}$, $4559 \pm 3504 \text{ mm h}^{-1}$, $5322 \pm 2120 \text{ mm h}^{-1}$, and $8303 \pm 1978 \text{ mm h}^{-1}$, respectively; SIRs were $63 \pm 44 \text{ mm h}^{-1}$, $82 \pm 57 \text{ mm h}^{-1}$, $167 \pm 72 \text{ mm h}^{-1}$, and $283 \pm 90 \text{ mm h}^{-1}$, respectively; CIs were $913 \pm 406 \text{ mm}$, $890 \pm 512 \text{ mm}$, $1470 \pm 387 \text{ mm}$, $2226 \pm 514 \text{ mm}$ for 1-, 2-, 3-, and 5-year-old eucalyptus plantations, respectively. The IIR, SIR, and CI increased with forest age (Fig. 5). The IIR, SIR, and CI of the 5-year-old eucalyptus plantation were the highest in four plantations ($p < 0.05$). The infiltration capacity of the 1- and 2-year-old eucalyptus plantations showed no significant difference, and were lower than other plantations.

The infiltration processes of preferential flow were fitted with the Horton model (Eq. 5), Kostikov model (Eq. 6), and Philip model (Eq. 7) in each plantation. As shown in Table 3, the variation coefficients of the three model parameters were higher than 10%, showing that the preferential flow had high variability. The R^2 of the three models were 0.77–0.98 (Horton model), 0.70–0.99 (Kostikov model), and 0.67–0.99 (Philip model), showing that the measured infiltration data accurately reflected the preferential flow infiltration process. The fitting parameters i_f (steady infiltration rate) and i_0 (initial infiltration rate) of the Horton model increased with forest age, similar to the measured results. The fitting parameter (A, steady infiltration rate) of the Philip model presented a large number of negative values because the model had a fixed attenuation coefficient of time (-0.5) with limiting the attenuation rate of preferential flow (Liu et al., 2011). Thus, the Philip model was unable to fit the stable infiltration rate of the preferential flow.

3.2. Root biomass of eucalyptus plantations of four ages

According to Eq. (2), the root biomass of the nine sample trees

ranged from 0.74 kg to 9.39 kg in four eucalyptus plantations (Fig. 6). The root biomass of the sample trees rapidly grew in the eucalyptus plantations, especially in the second and third years. The root biomass of the 5-year-old eucalyptus plantation was the highest in four plantations. The root biomass of the sample trees in 3-year-old eucalyptus plantation was significantly higher than those of the 1- and 2-year-old eucalyptus plantations ($p < 0.05$).

4. Discussions

4.1. Effects of eucalyptus plantations of different planting ages on preferential flow

The root systems of plants, including living and decaying roots, can produce root channels, which account for 70%–100% of the total of macropores in forest soils (Newman et al., 2004; Noguchi et al., 2017). Roots can change the size, connectivity, and distribution of soil pores, thereby affecting the development of preferential flow in forest soils (Liu et al., 2020). In our study, the correlations between preferential flow and root biomass and TP were examined, respectively.

In four forest-age eucalyptus plantations, IIR had a significantly positive correlation to root biomass ($p < 0.05$), and SIR and CI had highly significant and positive correlation to root biomass ($p < 0.01$, Table 4). The results showed that root biomass increased with forest age, and thus IIR, SIR, and CI increased. These findings were consistent with those of previous studies (Archer et al., 2016; Leung et al., 2017). Plant roots promote the development of soil macropores in two main ways: (1) The roots separate and squeeze the soil to form root channels by elongating, interspersing, and thickening (Liu et al., 2020); (2) when roots die, the decaying xylem forms a tubular root channel because the decay of bark takes more time than the decay of xylem (Gaiser, 1952). As forest age increases, increase in root biomass promotes the formation of plant root channels and increases macropore network density and continuity, thereby promoting preferential flow (Wu et al., 2021). Moreover, plant roots in the upper layer of soil form more preferential root channels than those in the subsoil during plant root decay (Jørgensen et al., 2002; Liu

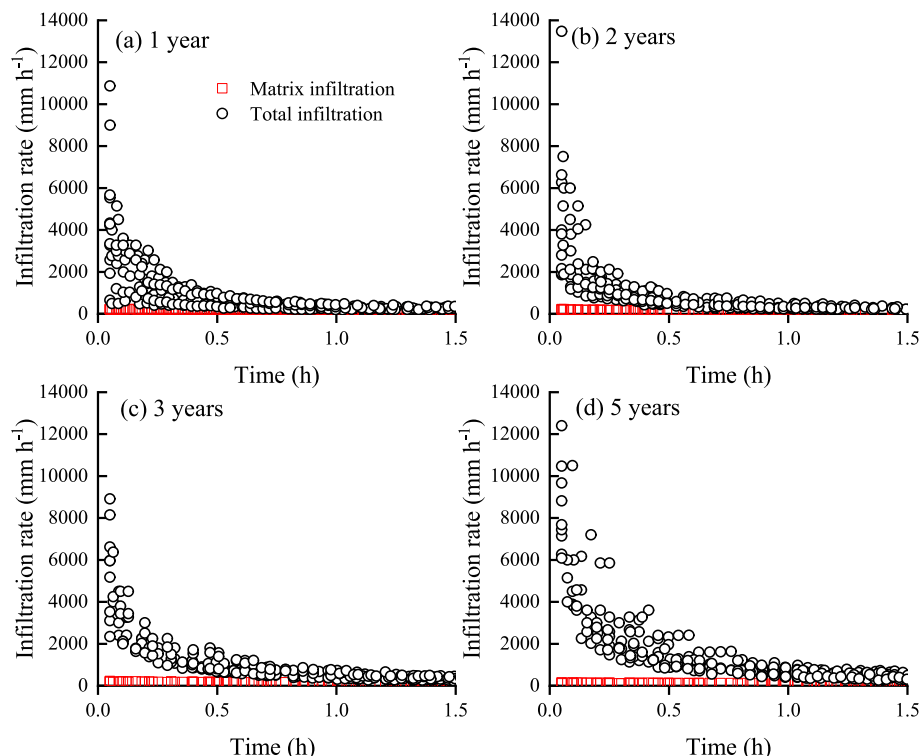


Fig. 3. Measurement results of total infiltration rate and matrix infiltration rate.

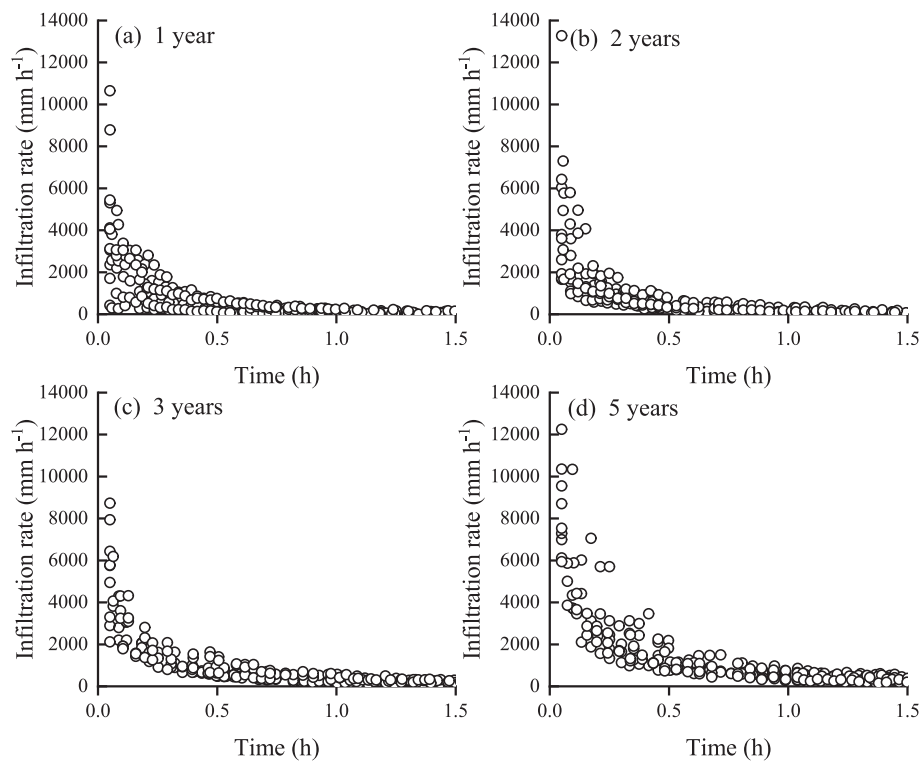


Fig. 4. Preferential flow infiltration process in four aged eucalyptus plantations.

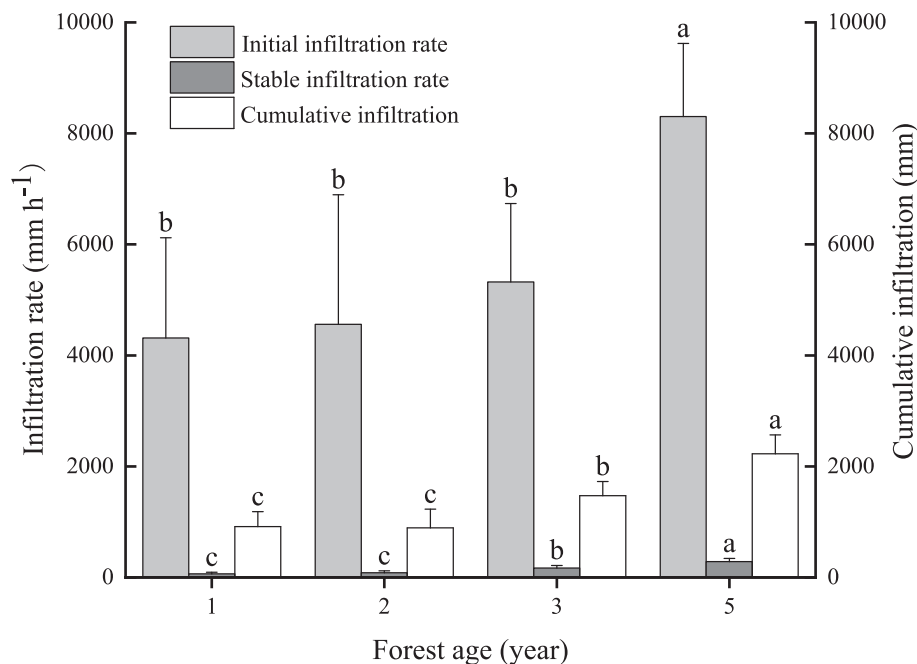


Fig. 5. Infiltration characteristics of preferential flow in four aged eucalyptus plantations. Error bars show 95% confidence intervals. Different lowercase letters indicate significant differences in characteristics of preferential flow among differently aged eucalyptus plantations at the 0.05 level.

et al., 2020), resulting in a high IIR. Owing to increase in root biomass with forest age, root decay increases, then leading to IIR increase.

However, the correlations between TP and IIR, SIR, and CI were all not significant ($p > 0.05$). The findings were inconsistent with some previous studies (Ren et al., 2016; Kalhor et al., 2018). Their studies focused on soil total infiltration (including matrix infiltration and preferential flow) rather than preferential flow. The macropores, which were the pores through preferential flow, accounted for only a small

proportion of soil (Beven and German, 2013; Moret and Arrue, 2007). So, preferential flow may be not significant with TP. However, other previous researches were consistent with our results (Shi et al., 2021; Jacka et al., 2021). Shi et al. (2021) emphasized that non-capillary porosity, that is macropores, plays a more important role in changes of infiltration than TP. The macropores had a significant effect on the preferential transport of water, the flow rate in which could be two to eight times higher than that in the soil matrix (Bouma, 1982; Wang et al.,

Table 3

Fitting parameters of preferential flow infiltration process by using three models in four aged eucalyptus plantations.

Forest age (year)	Regression models		Parameters	Sample tree number						Coefficient of variation	
				1	2	3	4	5	6		
1	Horton	$i = i_f + (i_0 - i_f)\exp(-nt)$	i_f	52.19	154.52	48.67	27.33	45.15	22.96	75.75%	
			i_0	5595.71	4913.55	21569.02	5025.43	368.31	7147.65	89.42%	
			n	4.40	5.25	16.72	7.89	1.43	7.79	65.79%	
			R^2	0.95	0.98	0.96	0.96	0.77	0.98	—	
	Kostiakov	$i = at^b$	a	446.65	384.17	156.97	165.77	143.70	209.37	47.50%	
			b	0.85	0.83	1.38	1.05	0.35	1.09	34.05%	
			R^2	0.88	0.94	0.99	0.95	0.70	0.94	—	
			Philip	$i = 0.5St^{-0.5} + A$	S	2998.44	2308.1	4775.63	1985.72	160.21	2868.07
	A	−1337.12			−934.32	−2711.89	−1039.08	64.68	−1577.49	66.02%	
	R^2	0.94			0.97	0.89	0.94	0.67	0.95	—	
	2	Horton			$i = i_f + (i_0 - i_f)\exp(-nt)$	i_f	223.88	94.43	24.99	83.53	32.30
			i_0	17901.90		8953.80	1795.46	2440.95	2053.08	9611.55	81.37%
n			11.28	8.48		5.03	6.16	4.40	10.14	33.92%	
R^2			0.91	0.98		0.94	0.98	0.95	0.97	—	
Kostiakov		$i = at^b$	a	362.42	254.26	136.81	157.33	163.29	249.65	35.30%	
			b	1.15	1.08	0.85	0.85	0.83	1.09	13.71%	
			R^2	0.95	0.96	0.97	0.94	0.92	0.99	—	
			Philip	$i = 0.5St^{-0.5} + A$	S	5469.42	3354.28	882.04	1056.67	1039.33	3179.54
A		−2822.51			−1761.29	−378.16	−448.59	−436.11	−1557.57	73.17%	
R^2		0.88			0.94	0.99	0.97	0.95	0.96	—	
3		Horton			$i = i_f + (i_0 - i_f)\exp(-nt)$	i_f	141.80	229.52	184.23	288.28	111.67
			i_0	3295.60		7567.50	7942.77	7823.09	3557.49	2217.32	44.71%
	n		3.40	5.34		9.20	5.13	4.25	2.23	44.28%	
	R^2		0.98	0.82		0.95	0.97	0.97	0.98	—	
	Kostiakov	$i = at^b$	a	441.23	522.30	337.51	627.27	320.49	540.20	23.72%	
			b	0.67	0.87	0.94	0.81	0.79	0.50	18.80%	
			R^2	0.88	0.91	0.97	0.97	0.95	0.91	—	
			Philip	$i = 0.5St^{-0.5} + A$	S	1718.52	3454.96	2770.4	3569.72	1784.34	1144.75
	A	−519.15			−1284.98	−1146.00	−1297.30	−666.20	−61.15	54.83%	
	R^2	0.92			0.89	0.95	0.98	0.99	0.91	—	
	5	Horton			$i = i_f + (i_0 - i_f)\exp(-nt)$	i_f	471.11	211.54	359.96	217.38	174.76
			i_0	6606.99		14710.82	14405.77	14938.08	6438.95	7850.93	35.92%
n			3.11	4.50		8.32	10.90	5.40	4.77	42.75%	
R^2			0.94	0.98		0.97	0.94	0.95	0.98	—	
Kostiakov		$i = at^b$	a	1051.37	1107.68	619.33	334.78	470.74	785.00	39.03%	
			b	0.65	0.85	0.94	1.11	0.86	0.77	16.46%	
			R^2	0.95	0.89	0.99	0.99	0.98	0.97	—	
			Philip	$i = 0.5St^{-0.5} + A$	S	3596.76	7456.94	5056.20	4448.78	2992.82	3873.67
A		−864.59			−3165.02	−2037.08	−2052.43	−1149.03	−1263.39	43.87%	
R^2		0.97			0.96	0.96	0.91	0.98	0.99	—	

Note: i is infiltration rate, mm h^{-1} ; t is infiltration time, h; i_f , i_0 , n are the parameters of the Horton model; a , b are the parameters of the Kostiakov model; S , A are the parameters of the Philip model; R^2 is the determination coefficient; All models were significant at the 0.001 level.

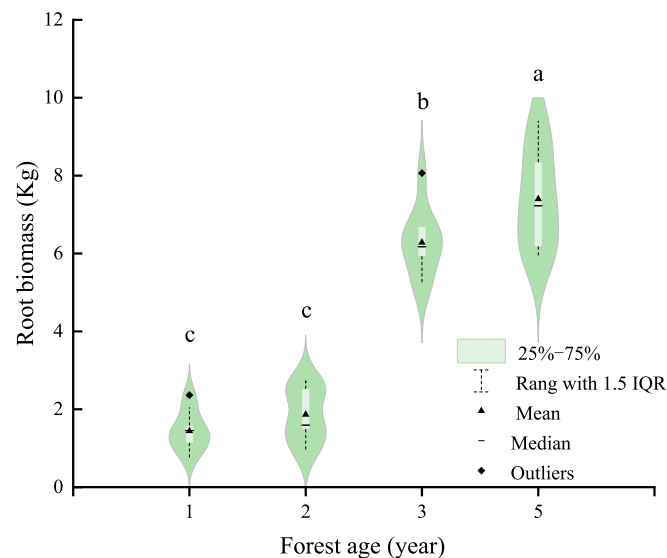


Fig. 6. Root biomass of the nine sample trees in four aged eucalyptus plantations. Different lowercase letters indicate significant differences in root biomass among differently aged eucalyptus plantations at the 0.05 level.

Table 4

Correlation between root biomass, total porosity and the infiltration characteristics of preferential flow in four aged eucalyptus plantations.

Soil matrix infiltration characteristic	Initial infiltration rate	Steady infiltration rate	Cumulative infiltration	Root biomass
Root biomass	0.422*	0.642**	0.655**	–
Total porosity	–0.159	–0.164	–0.110	–0.007
Dye coverage of soil profiles along tangents of inner ring	0.917**	0.841**	0.871**	0.661
Dye coverage of soil profiles along tangents of outer ring	0.868**	0.876**	0.883**	0.809*

Note: * means significant correlation at the 0.05 level ($p < 0.05$); ** means significant correlation at the 0.01 level ($p < 0.01$).

1994; Shipitalo and Gibbs, 2000; Zhang et al., 2019), and small changes could affect the mutation of the preferential flow (Cheng et al., 2018). Jacka et al. (2021) found that soil total infiltration and preferential flow in three plantations was not related to TP, while significantly influenced by pore size distribution and connectivity. In summary, the preferential

flow of eucalyptus plantations was mainly affected by its root, and not significantly affected by TP.

The vertical and horizontal growth of plant roots leads to the formation of a large number of vertical and horizontal channels of preferential flow in the soil (Ghestem et al. 2011; Jiang et al., 2018). After water infiltrates the soil, water flows along the lateral coarse roots, forming a rapid lateral preferential flow (Schwäzel et al., 2012; Guo et al., 2020). This phenomenon was consistent with our findings. As shown in Fig. 7a and b, some parts of the soil profiles along the tangents of the inner and outer rings were dyed, and a clear root channel from the inner ring to the outer ring was observed. The dyeing results showed that water infiltrating the inner ring flowed laterally outside of the inner ring through the root channels because of the lateral roots, and formed a lateral preferential flow (Fig. 7c, Zhang et al., 2019). The soil profile dyeing area was quantified, the dye coverages of the soil profiles along the tangents of the inner and outer rings were 1.13%–28.10% and 0.06%–4.73%. The dye coverages of the soil profiles along the tangents of the inner and outer rings increased with forest age. Moreover, correlation analysis showed that the dye coverage of the soil profiles along the tangents of inner and outer rings had an extremely significant and positive correlation to IIR, SIR, and CI ($p < 0.01$). And the dye coverage of the outer ring tangent was significantly and positively correlated with root biomass ($p < 0.05$). These results illustrated that the lateral preferential flow was controlled by the roots and increased with forest age, eventually increasing preferential flow. Lateral root pathways became deep and wide as forest age increased, promoting lateral preferential flow. Simultaneously, the lateral preferential flow expanded the potential soil volume accessible to surface water, enhancing preferential flow (Lange et al., 2009).

4.2. Unified model of preferential flow

Our results indicated that the Philip model is unsuitable for describing the SIR of preferential flow because the fitting parameter (A , steady infiltration rate) of the Philip model presented a large number of negative values, similar to the results of the previous studies (Zhang et al., 2017b; Santra et al., 2021). The measured IIR, SIR, and CI of the six sample trees in each plantation were individually compared with those fitted using the Horton and Kostiakov model and then the appropriate model was selected. As shown in Fig. 8a, the IIR estimated by fitting the results of the Kostiakov model was extremely high and 6.24 times the measured IIR. Given that the Kostiakov model is a power function, when the time is 0, the IIR tended to have infinite values.

The IIR estimated with the fitting results of the Horton model was closer to the measured values. The SIR estimated by fitting the results of the Horton and Kostiakov models were 1.07 and 1.72 times the measured values, respectively (Fig. 8b). The CI values estimated by the fitting results of the Horton model were 0.93 times the measured values, but the CI estimated by the fitting results of the Kostiakov model was 2.10 times the measured values (Fig. 8c). In summary, the IIR, SIR and CI predicted by the Horton model were closer to the measured values than those predicted by the Kostiakov model. Wang et al. (2017) showed that the fitting results of the three-parameter model (Horton model) were more accurate than those of the two-parameter model (Kostiakov and Philip models). Additional parameters adjust three-parameter models and improve their accuracy by increasing the stability of the models and reducing interference and bias (Duan et al., 2010; Wang et al., 2017). Hence, the Horton model was selected as the basis of the unified model for the preferential flow in short-rotation eucalyptus plantations.

Root biomass was positively and significantly correlated with i_f ($p < 0.01$) and i_0 ($p < 0.05$), respectively, which were the parameters of the Horton model. The effects of root biomass on preferential flow in eucalyptus plantations were considered in the regression analysis for predicting the parameters i_f and i_0 (Table 5). The parameter n is the infiltration decay factor (Haghighi et al., 2010) and significantly positively correlated with the difference between initial and steady infiltration rates ($i_0 - i_f$, $p < 0.01$). Thus, the parameters i_f and i_0 were used as independent variables in regression analysis for predicting the parameter n (Table 5). A unified model (Eq. 14) predicting preferential flow in short-rotation eucalyptus plantations was obtained.

$$i = (27.84W_R + 29.13) + (217.92W_R^2 - 1356.62W_R + 5341.26)\exp[-(0.11W_R^2 - 0.85W_R + 5.90)t] \quad (14)$$

where, i represents infiltration rate, mm h^{-1} ; t is infiltration time, h; and W_R denotes root biomass, kg.

The measurement results of the other three sample trees in each plantation were used in verifying and evaluating the effects of the unified model. The measured preferential flow infiltration process in the three sample trees were compared with the values predicted by the unified model. As shown in Fig. 9, the fitting coefficients between the measured and predicted values of the four eucalyptus plantations were close to 1, indicating that the unified model had a good predictive effect on the preferential flow the short-rotation eucalyptus plantations. The good prediction of unified model further showed that roots had a considerable effect on preferential flow in eucalyptus plantations.

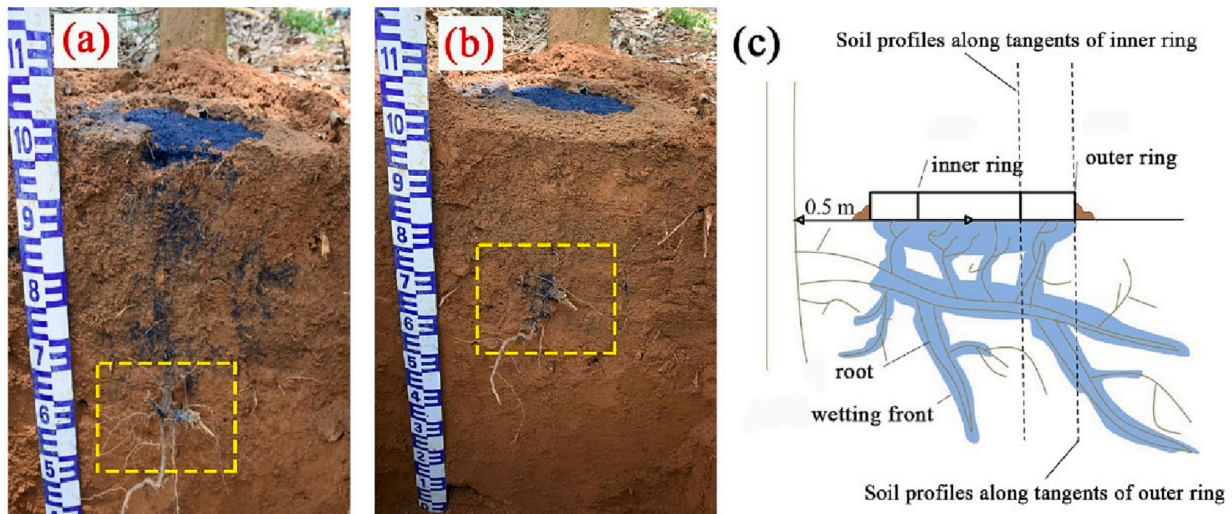


Fig. 7. Soil profiles and diagram of lateral preferential flow. (a) Soil profile along the tangents of the inner ring; (b) Soil profile along the tangents of the outer ring; (c) Diagram of preferential flow lateral transport.

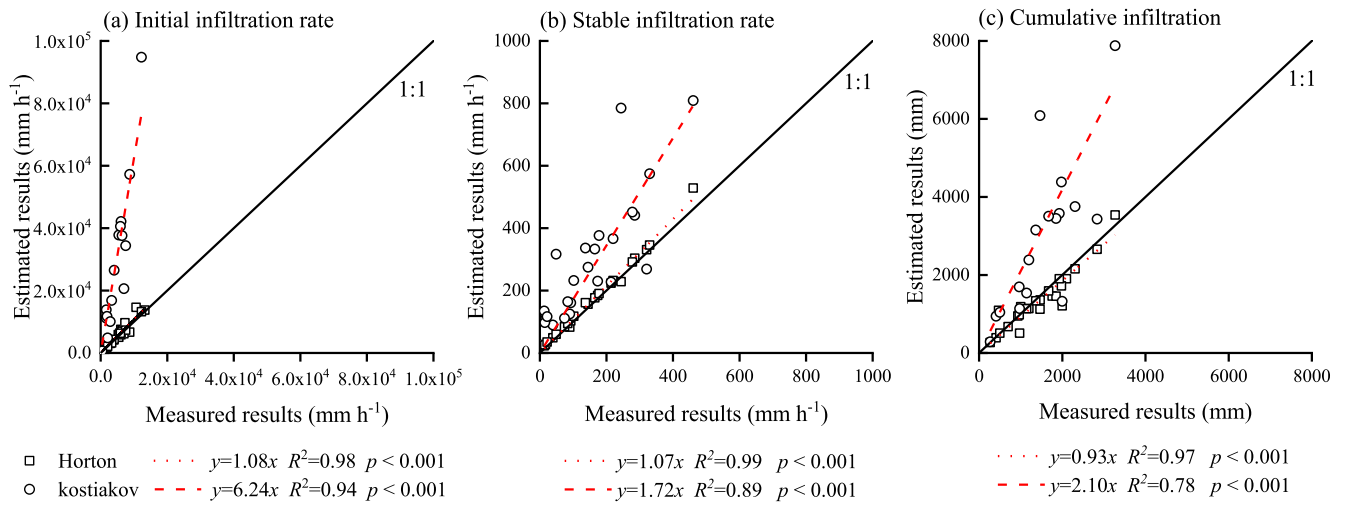


Fig. 8. Comparison between measured infiltration characteristics and infiltration characteristics estimated by the Horton and Kostiakov models.

Table 5

Regression equation of parameters of the Horton model.

Parameter	Regression equation	R^2	p
i_f	$i_f = 27.84W_R + 29.13$	0.52	< 0.001
i_o	$i_o = 217.92W_R^2 - 1328.43W_R + 5370.40$	0.54	0.003
n	$n = -0.007 i_f + 0.0005 i_o + 3.45$	0.65	< 0.001

Note: i_f , i_o , and n are parameters of the Horton model; W_R is the root biomass, Kg.

5. Conclusions

In this study, total infiltration and matrix infiltration in adjacent areas were measured by using a surface-positioned double-ring infiltrometer. The preferential flow infiltration processes in four aged (1, 2, 3 and 5 years) eucalyptus plantations were determined by calculating the differences between the processes of total infiltration and matrix infiltration. The results showed that IIR, SIR, and CI increased during short

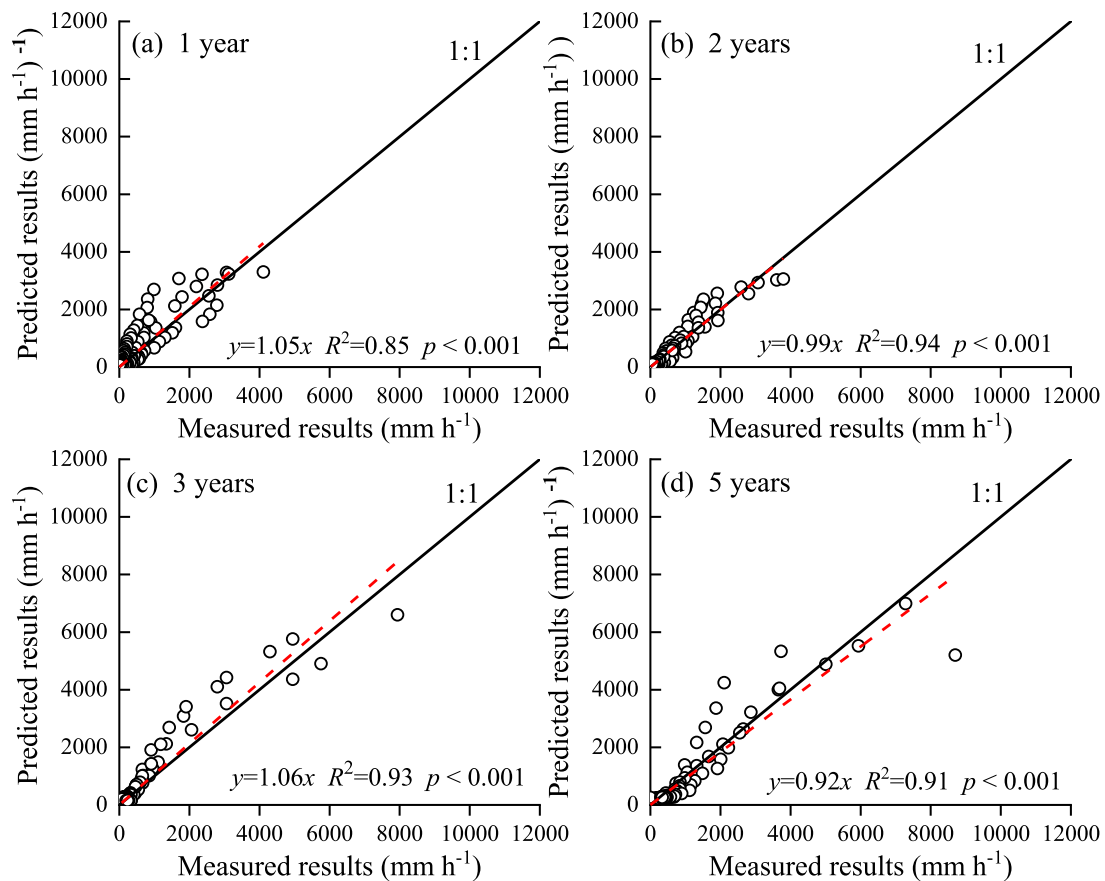


Fig. 9. Comparison of the infiltration processes of preferential flow predicted by the unified model and those measured by the field experiments.

rotation of eucalyptus plantation. The root biomass was significantly and positively correlated with IIR ($p < 0.05$), SIR ($p < 0.01$) and CI ($p < 0.01$). But, the correlations between TP and IIR, SIR, and CI were all not significant ($p > 0.05$). Preferential flow increased with root biomass mainly because of root-induced lateral preferential flow. In the three infiltration models, the Horton model had a higher stability and accuracy than the other two-parameter models (Kostiakov model and Philip model). Regression analysis was used in determining the parameters of the Horton model with considering root biomass, and then the unified model of preferential flow in short-rotation eucalyptus plantations was constructed. The unified model and correlation results demonstrated the significant effect of root biomass on preferential flow of eucalyptus plantations. Our results revealed a theoretical basis for improving the efficiency of water resource management in short-rotation eucalyptus plantations.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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